CLINICAL · APPROACH TO

Approach to difficulty swallowing

Difficulty swallowing is never a trivial symptom. It can mean a cancer of the pharynx or oesophagus, a neurological disease, a pharyngeal pouch or a motility disorder, and it carries the risks of aspiration pneumonia, malnutrition and dehydration. The approach rests on three questions: is the problem in the mouth and pharynx (*oropharyngeal*) or in the oesophagus; is it structural or neuromuscular; and are there features of malignancy that demand urgent endoscopy?

AT A GLANCE

- Coughing and nasal regurgitation: oropharyngeal (neurological).
- Solids then liquids, weight loss: cancer.
- Liquids and solids from the start: achalasia.
- Regurgitation of old food, gurgling: pharyngeal pouch.

1 CHAPTER ONE Where is the problem?

Oropharyngeal or oesophageal.

1.1 Two patterns

Oropharyngeal dysphagia is difficulty starting a swallow: food sticks in the throat at once, the patient coughs or chokes on swallowing, food or fluid comes back through the nose, the voice is wet, and repeated swallows are needed. It is usually neuromuscular (stroke, Parkinson's disease, motor neurone disease, myasthenia, myopathy) or caused by a pharyngeal pouch, a cricopharyngeal bar, head and neck cancer or its treatment.

Oesophageal dysphagia is a sensation of food sticking after swallowing, usually felt behind the sternum. Difficulty with **solids** that progresses to liquids suggests a mechanical obstruction: cancer if rapid with weight loss, a peptic stricture if slow with reflux, a ring or web if intermittent. Difficulty with **solids and liquids from the start** suggests a motility disorder such as achalasia. Patients localise the level poorly: a lower oesophageal obstruction can be felt in the neck.

RED FLAGS IN DYSPHAGIA

- **Progressive dysphagia, weight loss, odynophagia, haematemesis or anaemia:** upper GI endoscopy urgently, at any age.
- **Dysphagia with hoarseness, otalgia, a neck lump or a smoking and drinking history:** head and neck cancer.
- **Total dysphagia with inability to swallow saliva:** food bolus obstruction — emergency.
- **Coughing on swallowing, chest infections and weight loss:** aspiration; assess safety of oral intake.
- **New neurological signs** (ptosis, fatigable weakness, fasciculation, dysarthria): neurological cause.

CHAPTER TWO

2 History and examination

Detail matters.

2.1 History

Ask what is difficult (solids, liquids, tablets, saliva), where food sticks, and how long the problem has lasted and whether it is getting worse. Ask about coughing and choking, nasal regurgitation, a wet voice, regurgitation of undigested food (often hours later, with halitosis and gurgling in a pouch), chest infections, heartburn, weight loss, pain, voice change and neurological symptoms. Review medications that cause oesophagitis (doxycycline, bisphosphonates, potassium, NSAIDs) or dry the mouth. The **EAT-10** questionnaire screens for and measures dysphagia.

2.2 Examination

Examine the mouth, dentition, tongue movement and palate, the cranial nerves, the neck (nodes, thyroid, a pouch that gurgles), and the larynx and pharynx with a flexible endoscope, looking for pooling of saliva in the piriform fossae (a sign of obstruction or poor clearance), lesions and vocal fold movement. Observe a trial swallow of water when safe, and look for general signs of neurological disease.

3 CHAPTER THREE Investigation

Choose by the question.

SWALLOWING INVESTIGATIONS

TEST	WHAT IT SHOWS	USE
Flexible endoscopy in clinic	Mucosa, pooling, cord movement	First test in ENT
FEES (fibreoptic endoscopic evaluation of swallowing)	Pharyngeal phase, residue, penetration and aspiration	Oropharyngeal dysphagia, at the bedside
Videofluoroscopy	All phases of the swallow, aspiration, cricopharyngeal opening	Oropharyngeal dysphagia
Barium swallow	Pouch, webs, rings, strictures, motility	Structural and pouch assessment
Upper GI endoscopy (OGD)	Oesophageal mucosa, biopsy, dilatation	Oesophageal dysphagia, red flags
High-resolution manometry	Oesophageal motility (Chicago classification)	Achalasia, motility disorders
CT neck and chest	Malignancy and staging	Suspected cancer

In eosinophilic oesophagitis, biopsies are taken even when the oesophagus looks normal.

4 CHAPTER FOUR Causes

Structural, neuromuscular, oesophageal.

4.1 Pharyngeal pouch

A **pharyngeal pouch (Zenker's diverticulum)** is a pulsion diverticulum through **Killian's dehiscence**, between the thyropharyngeus and cricopharyngeus parts of the inferior constrictor, with failure of cricopharyngeal relaxation. It affects older adults and causes dysphagia, regurgitation of undigested food, halitosis, gurgling, aspiration and weight loss. A barium swallow confirms it. Symptomatic pouches are

treated by **endoscopic stapling or laser diverticulotomy**, which divides the wall between pouch and oesophagus together with the cricopharyngeus, or by open excision or inversion with cricopharyngeal myotomy for small or difficult pouches.

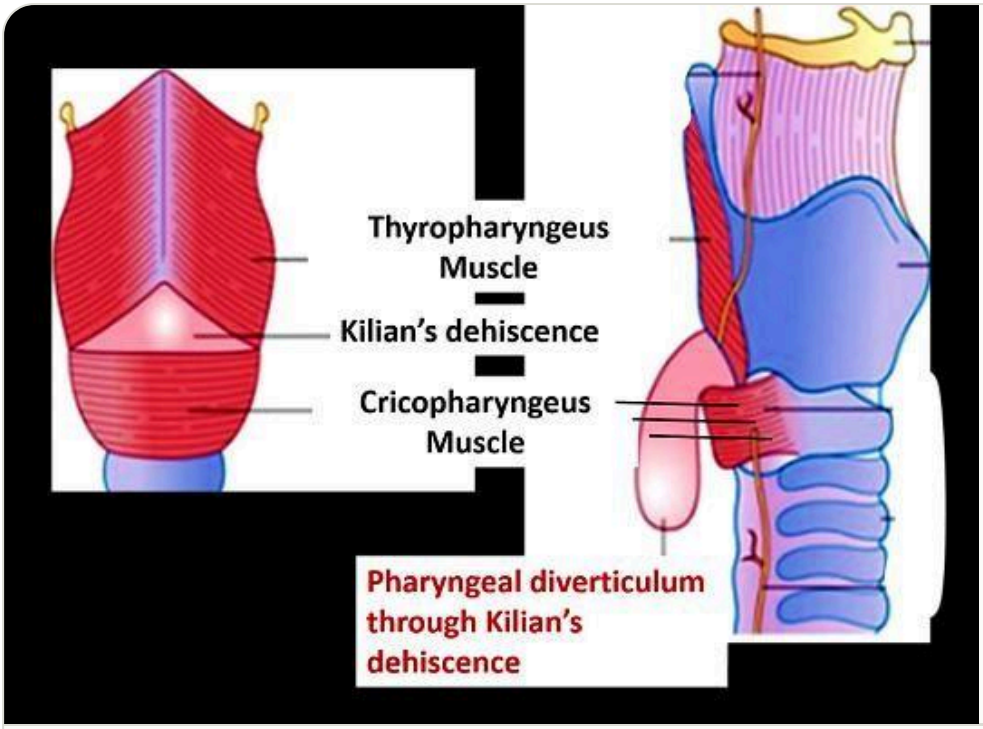


FIGURE 1 Killian's dehiscence and a pharyngeal pouch.
Anatomy atlas · ENT Steps p. 198

4.2 Other structural causes

Cricopharyngeal dysfunction without a pouch may respond to dilatation, botulinum toxin or myotomy. **Post-cricoid webs** with iron-deficiency anaemia form the Plummer–Vinson (Paterson–Brown-Kelly) syndrome, a risk factor for post-cricoid cancer. **Cervical osteophytes**, an enlarged thyroid and head and neck cancer, or the fibrosis that follows its treatment by radiotherapy, also narrow the pharynx.

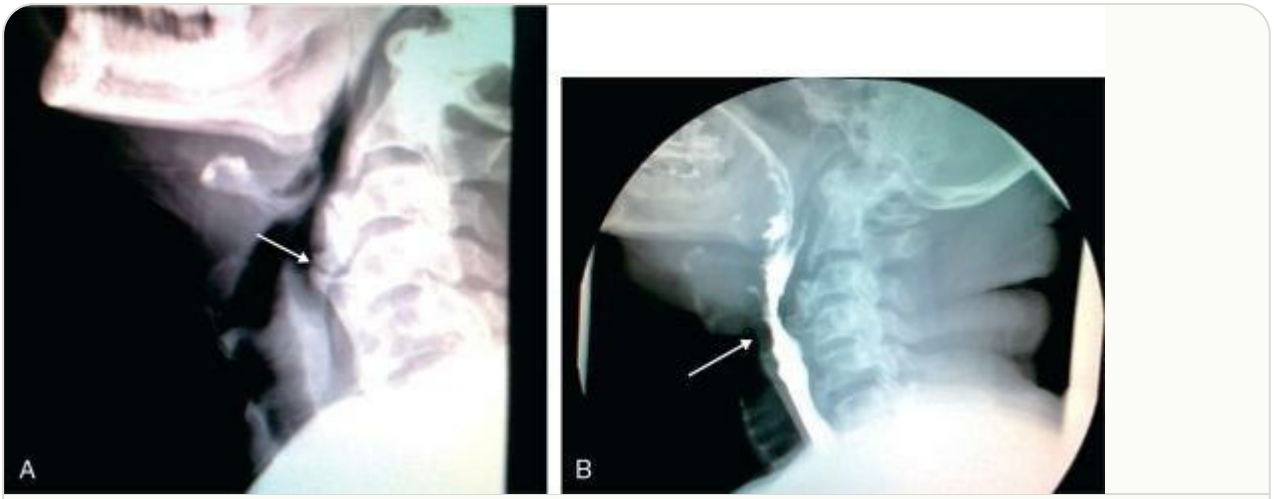


FIGURE 2 Large cervical osteophytes pressing on the oesophagus on barium swallow.
Dhingra, fig. 69.1

4.3 Neuromuscular causes

Stroke is the commonest cause of oropharyngeal dysphagia; most patients improve in the first weeks. Progressive neurological diseases (Parkinson's disease, motor neurone disease, multiple sclerosis, myasthenia gravis, inclusion body myositis, oculopharyngeal muscular dystrophy) need neurological diagnosis and long-term swallow management.

4.4 Oesophageal causes

Oesophageal cancer is the cause not to miss. **Peptic strictures** follow long-standing reflux. **Schatzki rings** cause intermittent food bolus obstruction. **Achalasia** is a failure of lower oesophageal sphincter relaxation with absent peristalsis, treated with pneumatic dilatation, Heller myotomy or peroral endoscopic myotomy. **Eosinophilic oesophagitis** typically affects young atopic men, presents with food bolus obstruction, and responds to proton pump inhibitors, topical steroids and dietary elimination.

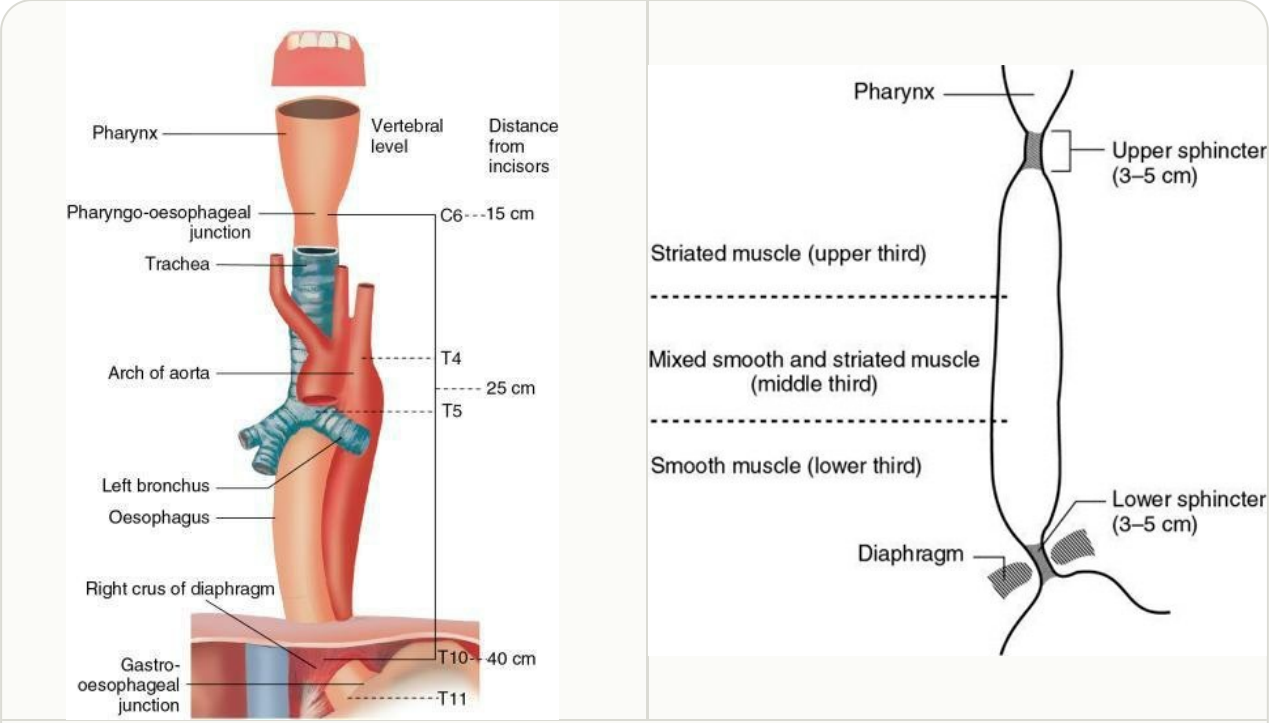


FIGURE 3 The oesophagus and its normal constrictions, with distances from the upper incisors.
Dhingra, figs 67.1 and 67.2

5 CHAPTER FIVE Management

Safety, cause, function.

5.1 Principles

Every patient needs an assessment of whether oral intake is safe, by a **speech and language therapist**, who can advise on texture modification, postures (chin tuck, head turn), exercises and compensatory manoeuvres. Nutrition and hydration are protected, by nasogastric or gastrostomy feeding when necessary. The cause is then treated. A **food bolus obstruction** in the oesophagus needs endoscopic removal if it does not pass, and every patient needs subsequent endoscopy to exclude a stricture, eosinophilic oesophagitis or cancer.

DIFFICULTY SWALLOWING

1 Red flags?

→ **progressive, weight loss, odynophagia, anaemia**

urgent OGD

→ **hoarseness, otalgia, neck lump**
head and neck examination and imaging

→ **cannot swallow saliva**
food bolus — urgent endoscopy

2 Oropharyngeal or oesophageal?

→ **coughing, nasal regurgitation, wet voice**
oropharyngeal, step 3

→ **food sticks after swallowing**
oesophageal — OGD, then barium or manometry

3 Oropharyngeal

→ flexible endoscopy, FEES or videofluoroscopy

→ **regurgitation and gurgling**
barium swallow for pouch — endoscopic stapling

→ **neurological signs**
neurology, speech therapy

4 Unsafe swallow

→ speech therapy, texture changes, postures

→ non-oral feeding if needed

CHAPTER SIX

6 For the examination

What the examiner wants to hear.

6.1 Viva essentials

WHAT THE EXAMINER IS LOOKING FOR

- **Oropharyngeal versus oesophageal dysphagia;** solids versus liquids.
- **Red flags** and urgent endoscopy.
- **FEES versus videofluoroscopy.**
- **Pharyngeal pouch:** Killian's dehiscence, presentation, endoscopic stapling and its limits.
- **Plummer–Vinson syndrome, achalasia and eosinophilic oesophagitis.**

6.2 Common pitfalls

COMMON PITFALLS

- Attributing progressive dysphagia to reflux without endoscopy.
- Performing a rigid oesophagoscopy blind in a patient with a pouch, risking perforation.
- Forgetting a swallow safety assessment before restarting oral intake.

Written for this app from Oxford Textbook of Otolaryngology ch. 48, 80, Oxford Specialist Handbook ch. 3, Oxford Handbook of ENT ch. 3, Dhingra ch. 67, 68. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

On imaging

The imaging of this topic: key findings, differentials and representative scans.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Zenker Diverticulum (Pharyngeal Pouch)

IMAGING FINDINGS

- Outpouching from the posterior hypopharynx above the cricopharyngeus on barium swallow
- Air–fluid level behind the oesophagus on CT
- Residual barium after the swallow

TOP DIFFERENTIALS

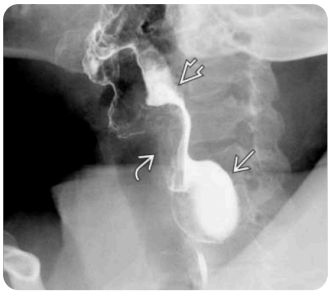
- Lateral pharyngeal or Killian–Jamieson diverticulum
- Paratracheal air cyst

CHECKLIST

- Size and neck of the pouch
- Aspiration on the swallow



Barium swallow, anteroposterior view. After swallowing contrast, some contrast has remained in the hypopharynx (1) due to physiologic…



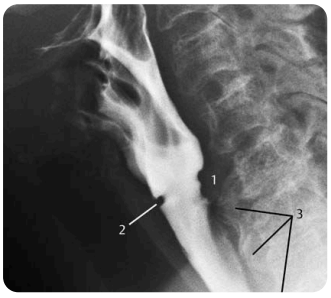
Barium esophagram demonstrates a large diverticulum with retained layering contrast from the posterior lateral junction of the hypopharynx…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 258.

PHARYNX, TONSILS & OESOPHAGUS · INFECTION & INFLAMMATION

Esophageal Passage Disorders

- Patients with removable dentures may lack oral sensory input regarding the size of their food. This may also happen with sharp foreign bodies, which may cause perforations of the esophagus.
- Anatomic structures such as the aortic arch or an aberrant right esophageal artery are demonstrated as persistent impressions without stasis of contrast.

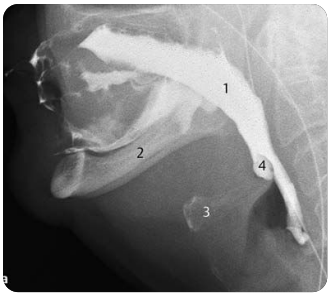


Barium swallow, lateral view. Problems with esophageal passage can be a result of a variety of etiologies occurring either singly or in…

LARYNX & TRACHEA · INFECTION & INFLAMMATION

Aspiration

- Occasionally aspiration is caused by mechanical failure of the epiglottis to close because of the presence of tumor, but most cases are due to neurologic (sensory) disorders without any visible pathology apart from the aspiration. Beware of aspiration pneumonia.
- The above-mentioned pathologies are partly demonstrated in other sections.



First stage: the barium contrast (1) just reaches the esophagus. For orientation note the mandible (2), hyoid (3), and vallecula (4)....